

maeko flock

100% FREE

A hand-painted starling murmuration for Unity 6 (URP). Realistic painterly style, living flock behavior - swirls, hovers, panics, dive attacks, and patrol routes - with birds that arrive from the horizon. One drop-in component.

[Documentation & Setup Guide](#)

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SECTION 1

Introduction

Maeko Flock is a living starling murmuration for Unity 6 and the Universal Render Pipeline. The bird is modeled and painted by hand in a realistic painterly style: a true European starling, iridescent greens and purples, pale speckles, feathering laid down with visible brushwork. It reads as a real bird up close and as living smoke in the hundreds.

It is completely free. No locked features, no upsell inside: the full flock intelligence ships in this package.

At a glance

- Hand-painted realistic painterly starling, 172 triangles, flock ready
- One component murmuration: separation, alignment, cohesion, and a liquid blob that stretches with speed
- Four modes: Calm Swirl, Hover, Panic, and Dive Attack, switchable live or from script
- Figure eight, racetrack, and circle swirl patterns, plus unlimited patrol routes
- Birds arrive from the horizon and depart the same way: trigger zones, timers, or script
- Life built in: drifting scouts, sharp twitches, wind sway, ball breathing, head stabilization
- Bounding flight animation: flap bursts with natural glide pulses, speed synced per bird
- 3D chirp audio slot: drop in any loop and it pans, fades, and dopplers from the flock's center
- Two demo scenes and a clean, documented, warning free codebase

SECTION 2

Quick Start

10 seconds

- Create an empty GameObject and add the Maeko Flock component.
- Drag the StarlingBird prefab into Bird Prefab.
- Press Play. A murmuration gathers around the anchor. Done.

Good to know

- Move the Anchor transform at runtime and the whole flock follows it. Dragging it during Play mode is the fastest way to feel the system.
- One MaekoFlock component is one flock. Drop several in a scene, each with its own anchor, for multiple murmurations. Two flocks of 100 cost less than one flock of 200.
- The flock stays above the ground automatically: birds never sink more than Floor Below Anchor units under the anchor.
- For 250+ birds, tick Enable GPU Instancing on the bird material.

SECTION 3

Demo Scenes

Both scenes are included in MaekoFlock/Demo. Open one and press Play.

Bird Showcase (MaekoFlockShowcase)

The starling up close on a slow turntable in a white studio. It auto-toggles between the two flight animations, with a note on what triggers each inside the flock, and a Switch Animation button.

Simulation Room (MaekoFlockSimRoom)

A locked camera on a full flock. Arrow buttons browse setups: figure eight, racetrack, hover, liquid stretch, drifters, panic, dive attack, patrol, and migration, each with a short description. Page 1 is Full Auto: it plays the entire tour by itself like a highlight reel.

Recording

Press H in either demo to hide the UI for clean screenshots and video.

SECTION 4

Flock Modes

Calm Swirl (default)

The murmuration. The ball traces the selected swirl pattern around the anchor, breathing, swaying, and twitching. Drifting scouts peel off ahead and beside the flock, then sprint home.

Hover

Parks the ball directly on the anchor, tighter and denser. Perfect over a landmark or for a story beat.

Panic

Call `TriggerPanic()` from any script, or use the component's right click menu. The flock scatters climbing hard, flaps fast, then settles back into the swirl on its own after `Panic Duration`.

Dive Attack

Assign any transform to Dive Target. The flock collapses into a tight orbit above it while up to `Max Divers` birds take turns diving at it and pulling up.

From code

- `flock.SetMode(MaekoFlock.FlockMode.Hover);`
- `flock.TriggerPanic();` or `flock.TriggerPanic(explosionPosition);`
- `flock.SpawnFlock();` / `flock.DespawnFlock();` / `flock.BirdCount`

SECTION 5

Settings Reference

Bird Spacing

Personal space per bird. The ball's size grows with bird count automatically so spacing stays constant: 500 birds make a visibly bigger ball than 100. This is the density dial.

Blob Tightness / Group Unity

How hard strays are pulled back in, and how strongly every bird matches the flock's shared direction. Unity is what makes one ball move as a single organism.

Swirl Pattern / Radius / Speed

The path in Calm Swirl: Circle, Figure Eight, or Racetrack. Smaller radius keeps the show tighter on the anchor.

Stretch

How much the ball elongates along its direction of travel, like liquid. 0 keeps a rigid sphere.

Breathe / Sway / Twitch

Slow expansion and contraction of the whole ball, wandering drift of its center, and sharp little per-bird dodges.

Drifter Chance / Drifter Time

How often lone scouts peel off to fly ahead, beside, or above the flock, and for how long, before sprinting back.

Glide Amount

How often birds pulse into a glide during level flight. 0 is almost pure flapping, 1 is frequent soaring. Dives always glide.

Obstacle Mask / Avoid Distance

Set the mask to your terrain and building layers and birds steer around them. Leave as Nothing to skip the physics cost entirely.

Floor Below Anchor

Birds never fly lower than this many units below the anchor.

SECTION 6

Patrol, Spawn & Despawn

Patrol routes

Assign 2 or more transforms to Patrol Points and the flock flies to each in order, then loops back to the first, forever, at Patrol Speed. The route draws in the Scene view when the component is selected. Leave the list empty and the flock stays home around its anchor.

Spawning

- On Start: the flock is always there (default).
- On Trigger Enter: add any collider to the flock object, tick Is Trigger, size it as your zone. When something tagged with Player Tag walks in, the flock appears.
- By Script: call `SpawnFlock()` from a quest, a door, anything.
- With Arrive From Far on, birds spawn strung out along a line Arrive From Distance away and stream in like a migration. Untick for instant spawn.

Despawning

- Never, After Seconds (lifetime timer), or After Patrol Loops (finish N circuits of the route, then leave).
- Departing birds stream toward the horizon, despawning out of sight. The chirp fades with them.
- A trigger spawned flock can re-trigger after it leaves, so a zone has birds every time the player returns.

SECTION 7

The Animator & Audio

The animator (already set up in the demo prefab)

The rig is Generic. The controller needs exactly two states and NO transitions between them: Maeko_Fly (default) and Maeko_Glide. The flock cross-fades them itself: flap speed follows each bird's airspeed, and glides fire on dives, descents, and natural bounding flight pulses (see Glide Amount).

- States are found by name OR by Blender's exported clip name (Object|Action), so states dragged straight from the FBX work as-is.
- Only Maeko_Fly is required. Without a glide state, birds simply always flap.
- If birds fly sideways, rotate the model child inside the bird prefab so the beak faces the blue Z axis.

Audio

Drop ANY looping bird-chatter clip into Chirp Clip and it plays as 3D sound from the center of the ball: it pans left and right as the flock crosses the listener, swells as it approaches, fades to silence at Chirp Max Distance, with a touch of doppler as it sweeps past. No clip is shipped, bring your favourite: freesound.org has excellent CC0 starling loops.

SECTION 8

Maeko Sky

The murmuration was built to fly through Maeko Sky, our painterly day and night atmosphere for URP: full day-night cycle, sun, moon and stars, and two hand-painted cloud styles with horizon-matched fog. There is no dependency in either direction. They simply share the same brush.

Find Maeko Sky on the Unity Asset Store, or see everything we make at maeko.studio.

SECTION 9

Support

Maeko Flock is 100% free, made with real care by a tiny studio. If something is broken or confusing, email us FIRST and we will fix it fast, usually within a day or two:

MaekoStudio@outlook.com

And if the flock earns a place in your project, an honest review on the store page genuinely helps more than you know. It is the main way a free asset gets seen.

Thank you for flying with us.